

Proposed Solution

Date	29 June 2026
Team ID	XXXXXX
Project Title	RISING WATERS
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The Rising Waters project aims to reduce the impact of floods through a real-time flood monitoring and early warning system. The solution combines weather information, water-level monitoring, and interactive maps to provide timely alerts to users. It improves disaster preparedness, minimizes property damage, and enhances public safety by delivering accurate notifications and evacuation guidance.

Project Overview	
Objective	To develop a real-time flood monitoring and early warning system that predicts flood risks, provides instant alerts, and helps people take timely action to reduce loss of life and property.
Scope	The project collects weather and water-level data, analyzes flood risks, displays flood-prone areas on maps, and sends emergency notifications through SMS, email, and the web application.
Problem Statement	
Description	Many flood-prone regions experience delayed warnings and lack real-time monitoring, leading to property damage, traffic disruptions, and risks to human life.

Impact	The proposed solution improves disaster preparedness, enables faster emergency response, reduces flood-related losses, and increases public awareness through timely alerts.
Proposed Solution	
Approach	Develop a web-based flood monitoring system that integrates real-time weather data, water-level information, and GPS mapping to predict flood risks and provide instant notifications to users.
Key Features	Real-time flood monitoring and alerts. Weather forecasting and water-level tracking. Interactive flood-risk maps using Google Maps. SMS and email emergency notifications. Safe evacuation route guidance.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Resource type	CPU	Intel Core i5 or equivalent
Memory	RAM	8 GB
Storage	Disk space	256 GB SSD
Software		

Frameworks	Backend Framework	Flask
Libraries	Python Libraries	Requests, pandas, numpy, folium
IDE	Development Tool	Visual Studio Code / PyCharm
Data		
Data	Source, size, format	OpenWeather API, Government Flood Data, Water-Level Sensor Data(JSON/CSV)